

DecodeLabs – Project 1: Data Cleaning and Preparation

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Project: Project 1 – Data Cleaning and Preparation

1. Objective :

The objective of this project is to clean and prepare the given dataset for further analysis. The cleaning process includes handling missing values, checking duplicate records, validating dates and numeric fields, maintaining consistency in text and category fields, and verifying calculated values.

2. Dataset Overview :

The dataset contains 1,200 records and 14 columns. It contains information related to orders, customers, products, quantities, prices, dates, coupon codes, and other relevant fields. The dataset was reviewed and cleaned to make it suitable for further analysis.

3. Data Cleaning Process:

3.1 Handling Missing Values :

The dataset was checked for missing values. A total of 309 missing values were identified in the **CouponCode** column. These missing values were replaced with **No Coupon** to maintain consistency and avoid blank entries.

3.2 Duplicate Records :

The **OrderID** column was checked for duplicate records. No duplicate OrderIDs were found in the dataset.

3.3 Date Validation :

The date-related fields were checked to ensure that the values were valid dates. All date values were found to be valid.

3.4 Numeric Data Validation:

Numeric columns such as quantity, unit price, and total price were checked to ensure that they contained appropriate numeric values.

3.5 Text and Category Validation :

Text and categorical columns were reviewed to check for consistency and ensure that the values were appropriately formatted.

3.6 TotalPrice Validation :

The **TotalPrice** values were validated against the calculated values using the relevant quantity and unit price information. The calculated totals were found to be consistent with the dataset.

4. Validation and Quality Checks :

4.1 Record Count Check :

After cleaning, the dataset contains 1,200 records and 14 columns.

4.2 Duplicate Check:

The **OrderID** column was checked again after cleaning, and no duplicate records were found.

4.3 Missing Value Check:

The missing **CouponCode** values were handled by replacing them with **No Coupon**. The cleaned dataset was checked to ensure that the missing values were properly addressed.

4.4 Data Format Check:

Date, numeric, text, and categorical fields were reviewed to ensure that the data was in the appropriate format.

4.5 Calculation Check: The **TotalPrice** column was checked against the calculated total using the relevant quantity and unit price values. The values were successfully validated.

5. Final Result :

After completing the data-cleaning and validation process, the dataset was successfully prepared for further analysis. The final dataset contains 1,200 records and 14 columns. Missing values were handled, duplicate records were checked, dates and numeric fields were validated, text and category fields were reviewed, and the **TotalPrice** calculations were verified.

6. Conclusion:

The data-cleaning process improved the quality, consistency, and reliability of the dataset. The cleaned dataset is now organized and ready for further data analysis and visualization.